

AI MEETING ASSISTANT

An AI-powered system for meeting transcription and structured meeting insights

1. Project Overview

AI Meeting Assistant is a web-based application designed to reduce the manual effort required to document meetings. Instead of asking users to listen to a complete recording and manually write notes, the system accepts a meeting recording and uses AI to transform the spoken content into structured and useful meeting information. The current interface provides an output-language setting and supports audio or video uploads, making the application suitable for common meeting recording formats.

The implemented workflow connects two main AI tasks: transcription of the meeting recording and analysis of the resulting text. The analysis stage can identify actionable information such as tasks, responsible people, deadlines, and unresolved questions. For example, the tested system successfully produced action items such as a presentation task assigned to Sara and a login-fix task assigned to Omar, while also identifying an open question about final deployment.

2. Problem Statement

Meetings often contain important decisions, assignments, deadlines, and follow-up questions, but these details can be difficult to recover from a long recording. Manual note-taking also requires continuous attention and may result in missing or incomplete information. The project addresses this problem by providing an automated pipeline that converts meeting recordings into structured information that is easier to review and act upon.

3. Project Objectives

Accept meeting audio or video recordings through a simple browser-based interface.

Generate a transcript from the meeting recording using an AI-based transcription workflow.

Analyze the transcript with a generative AI model.

Extract actionable tasks together with available owners and deadlines.

Identify open or unresolved questions that require follow-up.

Present the generated meeting information in a clear and user-friendly format.

Provide a practical foundation that can be extended with additional meeting-analysis features.



4. System Architecture and Workflow

The application follows a sequential AI-processing workflow. The user begins by selecting the desired output language and uploading a meeting recording. The recording is then passed to the transcription component. After transcription, the generated text is provided to the meeting-analysis component, which uses Google Gemini to interpret the conversation and return structured insights.

Stage	Component	Main Function
1	Streamlit UI	Collect language selection and meeting recording.
2	Transcription	Convert spoken meeting content into text.
3	LLM Analysis	Interpret the transcript and extract meeting insights.
4	Structured Output	Present action items and open questions clearly.

5. Technologies Used

Technology	Role	Reason for Use
Python	Core development language	Implements the application logic and AI-processing modules.
Streamlit	Web interface	Provides a lightweight interactive interface for uploading and reviewing meeting data.
Google Gemini API	Generative AI	Performs language understanding and structured meeting analysis.
python-dotenv	Configuration	Loads environment variables such as API credentials from the project environment.
Virtual Environment	Dependency isolation	Keeps project dependencies separated from the system Python installation.

6. Main Functionalities

Meeting recording upload: the interface accepts common audio and video formats and provides a file-size limit for uploads.

Output language selection: the user can select the language used for generated meeting insights while preserving the original transcript language.

AI transcription: the recording is converted into a text representation that can be analyzed by the application.

Action-item extraction: the system identifies tasks, available task owners, and deadlines.

Open-question extraction: unresolved questions that need clarification or follow-up are separated from completed tasks.

Structured presentation: AI results are displayed in a readable format instead of returning an unstructured block of generated text.

7. Example of Generated Output

Owner	Action Item	Deadline
Sara	Prepare the presentation slides for the client meeting	Not specified
Omar	Fix the login issue	Tomorrow
Not specified	Review the testing results	Before launch

The test also identified the open question: “Who will handle the final deployment?” This demonstrates the distinction between assigned work and information that still requires clarification.

8. Testing and Validation

Testing was performed incrementally during development. First, the individual action-item generation workflow was executed from the project environment. The test successfully imported the meeting-analysis modules, connected to the Gemini-based language-model client, and generated structured action items and open questions from a controlled meeting example. Dependency and environment validation: the application was configured in a dedicated Python virtual environment.

AI client validation: the Google GenAI client was successfully imported and used by the application.

Action-item validation: tasks were extracted with owners and deadlines when those details were available.

Missing-information handling: the system returned “Not specified” when an owner or deadline was not explicitly available.

Open-question validation: unresolved issues were separated from action items rather than being presented as completed tasks.

9. Current Results

The current prototype demonstrates the core end-to-end concept: a user can open the Streamlit application, upload a meeting recording, and use the AI pipeline to obtain structured meeting information. The successful action-item test indicates that the analysis layer can transform unstructured conversational content into practical task-oriented information.

10. Limitations

AI-generated transcripts and meeting insights can contain errors, especially when audio quality is poor or speakers overlap.

The accuracy of owners and deadlines depends on whether those details are explicitly present or inferable from the meeting content.

The current system should be treated as an assistant for meeting documentation, not as a replacement for human verification.

The prototype currently focuses on core transcription and meeting-insight generation rather than full enterprise meeting management.

11. Future Improvements

Add stronger transcript review and editing capabilities before final analysis.

Improve speaker identification and handling of multi-speaker conversations.

Add richer meeting summaries, key decisions, discussion topics, and participant information.

Provide exportable meeting reports such as PDF or DOCX.

Add persistent meeting history and search across previous meetings.

Introduce evaluation metrics and a larger test set to quantitatively measure transcription and extraction quality.

12. Conclusion

AI Meeting Assistant provides a practical application of Generative AI to a common productivity problem. By combining a simple Streamlit interface with an AI transcription and analysis workflow, the project converts meeting recordings into structured information that is easier to understand and act upon. The current prototype successfully demonstrates the core functionality and provides a strong foundation for future improvements in accuracy, reporting, storage, and meeting knowledge management.